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CUET UG

Business Studies

# Controlling

Unit VIII · Controlling · CUET UG 2027 Complete Mastery

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## Syllabus Alignment & Chapter Architecture

| Sub-Topic Domain | Official NTA CUET UG Unit VIII Syllabus Scope | High-Frequency Cognitive Skills Tested |
|---|---|---|
| **Concept & Importance** | Meaning, nature, and importance (accomplishing goals, judging standards, resource efficiency, employee motivation, order/discipline, coordination). | Identifying controlling importance dimensions from corporate scenarios; understanding cybernetic control. |
| **Planning vs Controlling** | Interdependence and relationship; forward-looking vs backward-looking dynamics; prescriptive vs evaluative nature. | Analyzing the reciprocal relationship: why controlling is blind without planning and planning is meaningless without control. |
| **The Controlling Process** | Five sequential steps: Setting Standards, Measuring Performance, Comparing with Standards, Analysing Deviations, Taking Corrective Action. | Chronological ordering of control stages; evaluating techniques of measurement (sample checking, personal observation). |
| **Deviation Analysis** | Critical Point Control (CPC) focusing on Key Result Areas (KRAs); Management by Exception (MBE) with tolerance limits. | Differentiating CPC (where deviation occurs) from MBE (magnitude of deviation); evaluating corporate exception scenarios. |
| **Control Techniques** | Traditional techniques (personal observation, statistical reports, BEP, budgetary control) vs. Modern techniques (ROI, Ratio analysis, Responsibility accounting, Management audit, PERT/CPM, MIS). | Classifying control tools into traditional vs modern; calculating BEP; identifying responsibility centers. |

**CUET HIGH-SCORING INSIGHT & KEY COGNITIVE TRAPS**

Controlling questions in CUET primarily test **Analysing Deviations (CPC vs. MBE)** and the **Planning-Controlling Relationship**. Master the core difference: **Critical Point Control (CPC)** focuses on **Key Result Areas (KRAs)** that impact enterprise survival (e.g., direct production cost vs stationery cost), whereas **Management by Exception (MBE)** focuses on **acceptable ranges of deviation (tolerance limits)**, escalating only significant deviations to top management ("An attempt to control everything results in controlling nothing"). Also, remember that Planning and Controlling are **both forward-looking and backward-looking**.

## Concept & Epistemological Nature of Controlling

Controlling is the managerial function of ensuring that actual organizational activities conform to planned activities. It involves measuring actual performance against predetermined standards, identifying deviations, analyzing their underlying causes, and taking prompt corrective action to ensure the realization of enterprise objectives.

## Core Epistemic Importance of Controlling

A well-designed managerial control system yields six vital institutional benefits highlighted in NCERT:

1. **Accomplishing Organisational Goals:** Controlling measures actual progress toward established targets, detects operational bottlenecks, and initiates corrective actions, steering the enterprise toward goal fulfillment.
2. **Judging Accuracy of Standards:** An effective control system enables managers to verify whether the predetermined standards are realistic and accurate. In dynamic environments, control reviews indicate whether standards must be revised upward or downward.
3. **Making Efficient Use of Resources:** By exercising vigilance over physical and financial operations, controlling prevents theft, pilferage, corruption, and resource wastage, maximizing total factor productivity.
4. **Improving Employee Motivation:** When employees know in advance what performance standards are expected of them and how their work will be evaluated, ambiguity is eliminated, boosting morale and driving high performance.
5. **Ensuring Order and Discipline:** Controlling creates an atmosphere of orderliness and strict institutional discipline by minimizing dishonest behavior, unauthorized absenteeism, and operational negligence.
6. **Facilitating Coordination in Action:** Predetermined departmental standards provide clear performance benchmarks across divisions, aligning interrelated departmental efforts toward unified corporate objectives.

## Operational Limitations of Controlling

Despite its paramount significance, controlling faces four inherent institutional constraints:

- **Difficulty in Setting Quantitative Standards:** In areas like employee morale, public relations, and corporate governance, establishing objective, quantitative standards is exceptionally difficult, introducing subjective managerial bias.
- **Little Control on External Environmental Factors:** An organization cannot control external variables such as government fiscal policies, macroeconomic technological shifts, foreign exchange rate fluctuations, or geopolitical conflicts.
- **Resistance from Operating Employees:** Control systems (such as closed-circuit cameras, biometric surveillance, or strict output tracking) are often perceived by employees as infringements on autonomy, provoking friction and resistance.
- **Costly Affair:** Installing and operating an advanced control apparatus (like ERP systems, statistical process control, and real-time sensor networks) requires substantial financial investment and managerial time, which small enterprises cannot afford.

# Interrelationship Between Planning and Controlling

Planning and controlling are widely recognized as the **inseparable Siamese twins of management**. Neither function can operate in isolation.

| Analytical Dimension | Planning | Controlling |
|---|---|---|
| **Epistemic Purpose** | Prescriptive: determines objectives and charts future courses of action. | Evaluative: assesses actual execution against planned benchmarks. |

| Analytical Dimension | Planning | Controlling |
|---|---|---|
| **Temporal Orientation** | Forward-looking (focuses on future scenarios); yet guided backward by past control feedback. | Backward-looking (post-mortem evaluation of past performance); yet forward-looking (corrective action for future cycles). |
| **Mutual Dependence** | Meaningless without controlling: plans remain wishful dreams without execution tracking. | Blind without planning: control cannot evaluate performance without benchmark standards. |

## The Circular Nature of Management Control

The management process is cyclical: Planning initiates the management sequence by establishing standards, organizing allocates resources, staffing provides personnel, directing guides execution, and controlling measures output against standards. The corrective feedback generated by controlling feeds directly into subsequent planning cycles, establishing a continuous cybernetic feedback loop.

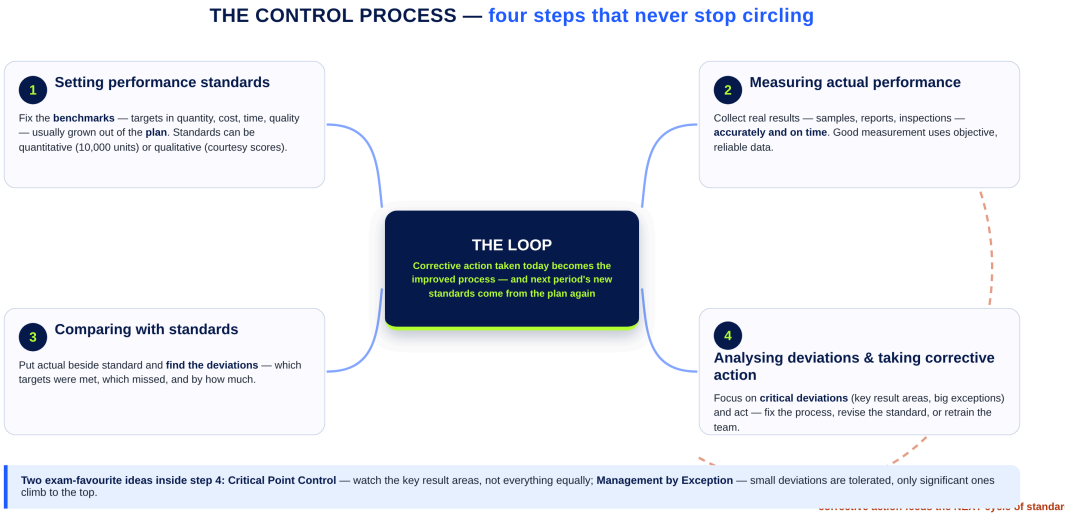



Figure 1: The Five Sequential Steps of the Managerial Controlling Process

# The Five Sequential Steps of the Controlling Process

The controlling process comprises five interconnected stages executed in a strict chronological sequence:

## Step 1: Setting Performance Standards

Standards serve as the criteria against which actual performance is measured.

- **Nature of Standards:** Standards can be **quantitative** (expressed in measurable numerical units such as 1,000 units produced per day, cost limit of ₹50 per unit, or 15% return on capital) or **qualitative** (goodwill, brand reputation, employee morale, customer satisfaction).
- **NCERT Guideline on Standards:** Whenever qualitative standards are used, managers must strive to establish observable proxy indicators (e.g., measuring customer goodwill via repeat purchase rates or customer complaints logged).
- **Flexibility:** Standards must not be rigid dogmas; they must adapt dynamically to technological shifts, market volatility, and changing environmental circumstances.

## Step 2: Measurement of Actual Performance

Actual operational performance must be measured objectively and reliably.

- **Key Techniques:**
  - **Personal Observation:** Supervisors observe operational floors firsthand, gathering qualitative insights, though subjective bias is possible.
  - **Sample Checking:** Used in mass production where inspecting every item is economically infeasible; statistically representative samples are audited.
  - **Performance Reports & Accounting Ratios:** Detailed quantitative summaries, variance logs, and financial ratios provide objective evaluation data.
- **Measurement Principle:** Performance should be measured in the same units in which standards are expressed to ensure direct and meaningful comparison.

## Step 3: Comparing Actual Performance with Standards

This step identifies the existence, magnitude, and direction of **deviations**.

- **Deviation Calculation:** Deviation = Actual Performance − Standard Performance.
- **Positive Deviation:** Actual performance exceeds standard (e.g., actual output 1,200 units vs. standard 1,000 units), signaling superior efficiency or under-pitched standards.
- **Negative Deviation:** Actual performance falls short of standard (e.g., actual output 850 units vs. standard 1,000 units), necessitating critical deviation analysis.

## Step 4: Analysing Deviations

Not all deviations require executive intervention. Minor fluctuations are natural in industrial and commercial processes. Managers apply two strategic principles to filter deviations:

### 1. Critical Point Control (CPC)

- **Principle:** It is neither economical nor practical to monitor every single organizational activity. Control must focus on **Key Result Areas (KRAs)** that are critical to the survival, profitability, and strategic success of the enterprise.
- **Logic:** If something goes wrong at a critical point, the entire enterprise suffers catastrophically.
- **NCERT Classic Example:** In a manufacturing enterprise, an increase of 5% in direct labor cost (a KRA that represents 60% of total product cost) requires immediate top-management intervention, whereas an increase of 15% in postal and stationery charges (which represents 0.1% of total cost) can be safely delegated or overlooked.

### 2. Management by Exception (MBE / Control by Exception)

- **Principle:** Based on the famous managerial maxim: **"An attempt to control everything results in controlling nothing."**
- **Logic:** Management establishes an **acceptable range of deviation (tolerance limit)**. Deviations that fall within this acceptable tolerance band are handled by lower-level operating staff. Only exceptional, major deviations that breach established tolerance limits are escalated to senior executives.
- **NCERT Classic Example:** A manufacturing tolerance band permits a variation of $\pm 2\%$ in production defects. If defects reach 1.5%, no executive alarm is raised. But if defects surge to 7%, the matter is immediately reported to the Chief Operating Officer.




Figure 2: Comparative Matrix: Critical Point Control (CPC) vs Management by Exception (MBE) & Control Techniques

### Root-Cause Diagnosis of Deviations

Once significant deviations are flagged via CPC or MBE, managers must identify root causes before prescribing solutions:

- **Defective Machinery:** Inadequate maintenance, worn-out tooling, or calibration drift → Overhaul or replacement.
- **Incompetent Personnel:** Skill deficiencies, inadequate training, or low motivation → Training or incentive restructuring.
- **Substandard Raw Materials:** Supplier non-compliance or poor quality control → Stricter procurement specifications.
- **Unrealistic Standards:** Flawed initial planning, impossible targets, or sudden macroeconomic shifts → Recalibration of standards.

## Step 5: Taking Corrective Action

The culmination of the controlling cycle is corrective action. If deviations are within tolerance limits, no corrective action is necessary. However, when deviations breach tolerance limits or threaten KRAs, immediate remedial intervention is mandatory.

| Nature of Operational Deviation | Identified Root Cause | Appropriate Corrective Action |
|---|---|---|
| Production output below target | Frequent machine breakdowns | Overhaul obsolete equipment or procure new machinery. |
| Defective finished goods output | Inferior grade raw materials | Change supplier; tighten input inspection protocols. |
| Project completion delay | Excessive bureaucratic sign-offs | Streamline administrative workflows and delegate authority. |
| Elevated unit manufacturing cost | High employee overtime hours | Improve scheduling, optimize shift patterns, and revise standards. |

# Traditional Techniques of Managerial Control

Traditional control techniques are time-tested tools that managers have utilized for decades:

## 1. Personal Observation

- **Mechanism:** The manager observes operational activities firsthand on the shop floor or in office departments.
- **Benefits:** Exerts psychological pressure on workers to perform diligently; provides unfiltered, experiential qualitative insights.
- **Limitations:** Extremely time-consuming; impractical for large-scale multi-plant enterprises; prone to subjective observer bias.

## 2. Statistical Reports & Data Visualizations

- **Mechanism:** Compiling quantitative operational data into tables, averages, percentages, ratios, and trend charts.
- **Benefits:** Highlights visual patterns, seasonal cycles, and historical trends that are invisible in raw transaction logs.
- **Limitations:** Historical in nature; reports past results rather than preventing errors in real time.

## 3. Break-Even Analysis (BEP)

Break-even analysis examines the dynamic cost-volume-profit relationship to determine the operational volume at which total revenues exactly equal total costs, resulting in zero profit and zero loss.

$$\text{BEP (in Units)} = \frac{\text{Fixed Costs}}{\text{Selling Price per Unit} - \text{Variable Cost per Unit}}$$

- **Contribution Margin:** Selling Price per Unit minus Variable Cost per Unit ("Price" - "Variable Cost").
- **Margin of Safety:** Actual sales volume minus Break-Even sales volume. A high margin of safety provides financial resilience against sales downturns.

## 4. Budgetary Control

Budgetary control is a comprehensive system of managerial control where all organizational operations are planned in advance in quantitative financial terms (budgets), and actual operational performance is continuously monitored against budget allocations to calculate variances.

### Major Classifications of Corporate Budgets

- **Sales Budget:** Forecast of expected sales volume and revenue by product line, sales territory, and customer segment.
- **Production Budget:** Planned output volume calculated as: Forecast Sales + Target Ending Inventory − Beginning Inventory.
- **Materials Budget:** Raw material input requirements, procurement schedules, and inventory buffer levels.
- **Cash Budget:** Projected cash inflows (customer collections, asset sales) and outflows (vendor payments, payroll, capital outlays), highlighting seasonal surplus or liquidity deficits.
- **Capital Expenditure Budget:** Planned investments in long-term fixed assets (machinery, real estate, software licenses).
- **Master Budget:** An integrated consolidation of all individual functional budgets into an overall corporate profit plan.

# Modern Techniques of Managerial Control

Modern techniques reflect sophisticated analytical and technological advances developed to handle complex, large-scale modern corporate enterprises:

## 1. Return on Investment (ROI)

ROI evaluates whether capital deployed in a business division or project is generating an adequate financial return.

$$\text{ROI} = \frac{\text{Net Profit Before Interest and Tax (EBIT)}}{\text{Total Capital Employed}} \times 100$$

- **DuPont Formula Breakdown:**

$$\text{ROI} = (\text{Net Profit Margin}) \times (\text{Capital Turnover Ratio}) = \left(\frac{\text{EBIT}}{\text{Sales}}\right) \times \left(\frac{\text{Sales}}{\text{Capital Employed}}\right)$$

- **Strategic Utility:** Enables corporate executives to compare performance across disparate divisional units regardless of absolute asset size; guides capital allocation toward highest-yielding business units.

## 2. Ratio Analysis

Ratio analysis evaluates enterprise financial health across four vital parameters:

- **Liquidity Ratios:** Current Ratio ($\frac{\text{Current Assets}}{\text{Current Liabilities}}$) and Quick Ratio; assess short-term solvency.
- **Solvency Ratios:** Debt-to-Equity Ratio and Interest Coverage Ratio; assess long-term solvency and capital structure risk.
- **Profitability Ratios:** Gross Profit Margin, Net Profit Margin, and Operating Ratio; measure earnings efficiency.
- **Turnover / Efficiency Ratios:** Inventory Turnover, Debtors Turnover, and Working Capital Turnover; evaluate asset utilization speed.

## 3. Responsibility Accounting

Responsibility accounting is a control paradigm where specific organizational segments are designated as **Responsibility Centers**, and an individual center manager is held strictly accountable for their performance.

| Responsibility Center | Managerial Authority | Primary Performance Metrics | Typical Corporate Unit |
|---|---|---|---|
| **Cost / Expense Center** | Incurs operational costs; no control over sales price or revenue. | Standard cost variances; operational efficiency; expense ceiling. | Manufacturing plant; maintenance department; legal division. |
| **Revenue Center** | Generates sales revenues; no authority over production manufacturing costs. | Gross sales volume; market share growth; sales price realization. | Regional sales office; digital e-commerce storefront. |
| **Profit Center** | Exercises authority over both revenues and operating costs. | Segment operating margin; gross margin; contribution to overheads. | Autonomous Strategic Business Unit (SBU); regional branch. |

| Responsibility Center | Managerial Authority | Primary Performance Metrics | Typical Corporate Unit |
|---|---|---|---|
| **Investment Center** | Controls operating revenues, costs, and capital investment assets. | Return on Investment (ROI); Economic Value Added (EVA). | Corporate subsidiary; international geographical division. |

## 4. Management Audit

Management audit is a systematic, independent, and comprehensive appraisal of the overall performance of the management of an organization.

- **Objectives:**
  - Evaluates managerial efficiency and the effectiveness of management policies.
  - Identifies systemic operational deficiencies in administration, leadership, and operational planning.
  - Recommends organizational restructuring and policy modifications to adapt to external market turbulence.
- **Key Difference from Financial Audit:** A financial audit checks accounting veracity and statutory compliance of financial books; a management audit evaluates the qualitative competence, strategic foresight, and overall effectiveness of executive management.

## 5. PERT and CPM Project Networks

Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM) are network-based scheduling and control tools utilized for complex, non-repetitive mega-projects (e.g., aerospace development, refinery construction, software release pipelines).

- **Origin & Nature:**
  - **CPM (Critical Path Method):** Developed by DuPont for chemical plant maintenance; **deterministic** in nature (activity durations are known with high certainty; focuses on cost-time trade-offs).
  - **PERT (Program Evaluation Review Technique):** Developed by the US Navy for the Polaris missile project; **probabilistic** in nature (used when activity durations are uncertain, utilizing three time estimates: Optimistic $t_o$, Most Likely $t_m$, and Pessimistic $t_p$).
- **The Critical Path:** The longest continuous sequence of dependent activities from project start to finish. Any delay in an activity lying on the critical path causes a corresponding delay in the entire project completion date. Activities on the critical path have **zero total float (zero slack time)**.

## 6. Management Information System (MIS)

MIS is a computer-based information system that captures, processes, stores, and delivers real-time operational data to managers for decision-making and control.

- **Data vs. Information:** Raw data consists of unorganized transactional facts (e.g., individual invoice timestamps); MIS transforms data into structured **information** (e.g., monthly regional product defect trends).
- **Benefits:** Eliminates data silos; provides executive dashboards with drill-down capabilities; enhances decision speed and precision.

# Advanced Dimensions of Quality and Cybernetic Control

Modern enterprises complement traditional controlling mechanisms with cybernetic control systems and quality frameworks that ensure continuous institutional self-correction.

## Feedforward, Concurrent, and Feedback Control Architecture

Depending on the timing of control intervention relative to the production process, control systems are categorized into three distinct operational architectures:

| Control Type | Temporal Stage | Primary Operational Focus | Illustrative Corporate Example |
|---|---|---|---|
| **Feedforward Control** | Before activity begins (Input stage) | Anticipatory: prevents anticipated problems by screening inputs. | Pre-flight aircraft maintenance checklists; testing raw material purity. |
| **Concurrent Control** | During activity execution (Throughput stage) | Real-time / Steering: adjusts operations as they are happening. | Direct supervisory observation on factory floor; computerized sensor monitoring. |
| **Feedback Control** | After activity completion (Output stage) | Historical / Evaluative: corrects deviations after output is generated. | Variance analysis; quarterly financial audit; customer feedback survey. |

## Total Quality Management (TQM) & Six Sigma Controls

In competitive global markets, controlling shifts from merely detecting defects to eliminating variance systematically:

- **Kaizen (Continuous Improvement):** A philosophy of relentless, small-scale incremental improvements involving every employee from senior management to shop floor workers.
- **Six Sigma (DMAIC Cycle):** A data-driven methodology striving for near-perfection (3.4 defects per million opportunities). It executes five phased steps: Define, Measure, Analyze, Improve, and Control.

# Comparative Analysis: Traditional vs Modern Control Techniques

| Evaluation Dimension | Traditional Control Techniques | Modern Control Techniques |
|---|---|---|
| **Technological Dependence** | Manual, document-driven, or basic statistical computation. | Highly computerized, automated, and algorithmic (MIS, ERP, CPM). |
| **Focus of Appraisal** | Historical financial data, physical output logs, and localized cost items. | Comprehensive organizational health, capital asset yields, and executive competence. |
| **Temporal Responsiveness** | Post-mortem review; feedback often arrives weeks after operational deviations occur. | Real-time feedback, interactive digital dashboards, and feedforward control. |
| **Organizational Scope** | Department-specific or localized operational units. | Enterprise-wide, cross-functional, and systemic integration. |

# Contemporary Corporate Case Studies in Managerial Control

## Case Study 1: Total Quality Control & The Andon Cord at Toyota

Under the Toyota Production System (TPS), every operating worker possesses the authority to pull the **Andon Cord**—halting the entire assembly line the instant a component defect is spotted.

- **Control Mechanism:** This represents **real-time concurrent feedback control**. Rather than letting defective assemblies pass to subsequent stations, deviations are arrested at their point of origin, preventing massive downstream compounding of defects.

## Case Study 2: Critical Point Control at Amazon Fulfillment

Amazon monitors thousands of operational metrics across its fulfillment centers. However, senior warehouse executives do not micromanage minor supplies costs. Instead, they focus relentlessly on two Key Result Areas (KRAs): **Click-to-Ship Cycle Time** and **Inventory Package Accuracy**.

- **Control Mechanism:** Applying **Critical Point Control**, Amazon ensures that customer satisfaction and warehouse throughput remain pristine, while automated systems manage peripheral operational variations within pre-calibrated tolerance bands.

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# Chapter Synthesis & Cognitive Mindmap

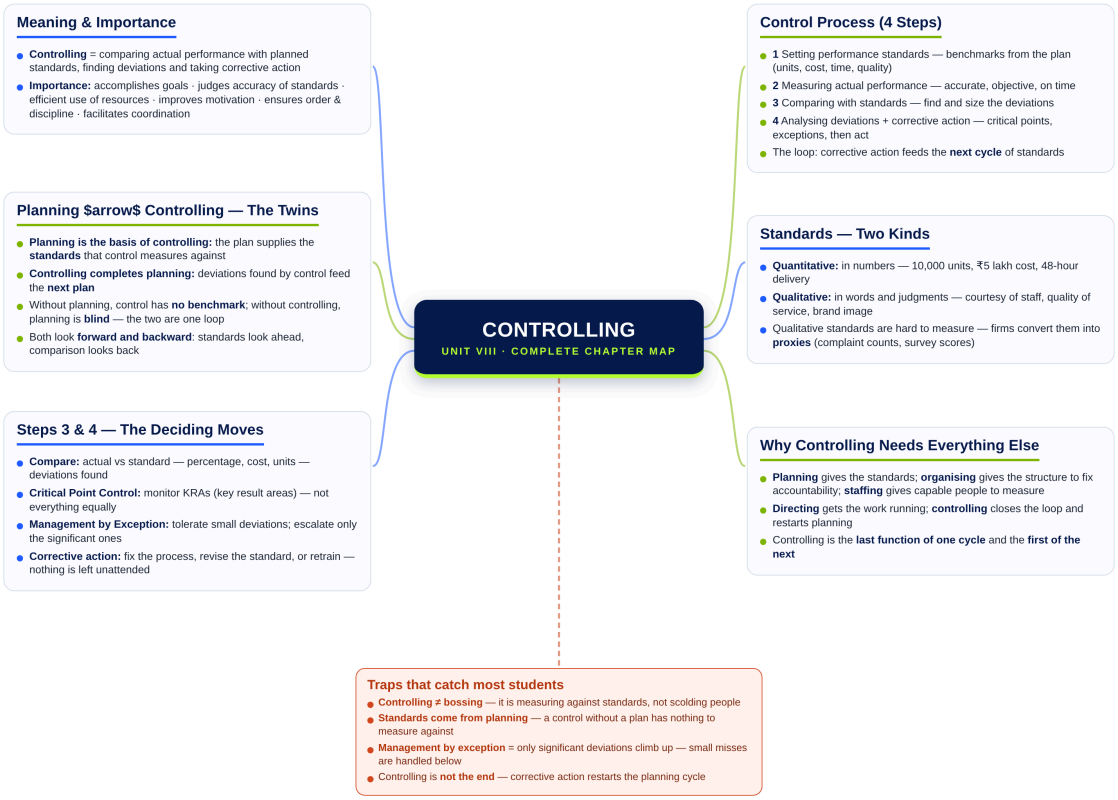



Figure 3: Hierarchical Knowledge Architecture: Controlling Concepts, Process, and Control Techniques

## CUET HIGH-YIELD MEMORY HOOKS & EPISTEMIC ASSOCIATIONS

- **Planning vs. Controlling:** Siamese twins of management; Planning is prescriptive (looking ahead), Controlling is evaluative (looking back), yet **both are forward-looking and backward-looking**.
- **Controlling is blind without planning:** Without standards established by planning, control has no benchmark for evaluation.
- **Planning is meaningless without controlling:** Plans remain theoretical exercises if execution is not verified and controlled.
- **The 5 Steps:** 1. Setting Standards → 2. Measuring Actual Performance → 3. Comparing with Standards → 4. Analysing Deviations → 5. Taking Corrective Action.
- **Critical Point Control (CPC):** Focuses on **Key Result Areas (KRAs)** where failure imperils the enterprise.
- **Management by Exception (MBE):** Focuses on **tolerance limits**. Deviations within the range are handled locally; only deviations beyond the range are escalated.
- **MBE Maxim:** "An attempt to control everything results in controlling nothing."
- **Traditional Techniques:** Personal observation, Statistical reports, Break-even analysis, Budgetary control.
- **Modern Techniques:** ROI, Ratio analysis, Responsibility accounting (Cost, Revenue, Profit, Investment centers), Management audit, PERT/CPM, MIS.

## MASTER STRATEGY CHEATSHEET — CONTROLLING

**1. Core Concept & Importance:**

- Bringing actual performance closer to planned performance.
- **Importance:** Accomplishing goals, judging standards, resource efficiency, employee motivation, order/discipline, coordination.
- **Limitations:** Difficulty in qualitative standards, little external control, employee resistance, costly affair.

**2. Planning-Controlling Relationship:**

- Planning is prescriptive; Controlling is evaluative.
- Both are forward-looking (corrective action / future goals) and backward-looking (audit / past control feedback).
- Controlling is blind without planning; Planning is meaningless without controlling.

**3. The 5-Step Controlling Process:**

- 1. Setting Standards (Quantitative vs Qualitative) → 2. Measuring Performance → 3. Comparing with Standards → 4. Analysing Deviations (CPC on KRAs; MBE on Tolerance Limits) → 5. Corrective Action.

**4. Control Techniques:**

- **Traditional:** Personal observation, Statistical reports, Break-even (BEP = FC / (P - VC)), Budgetary control.
- **Modern:** ROI (NP / CE × 100), Ratio analysis, Responsibility accounting (Cost, Revenue, Profit, Investment), Management audit, PERT/CPM, MIS.

## THE SEVEN FATAL CUET TRAPS — CONTROLLING

- **Trap 1: CPC vs. MBE:** CPC focuses on **WHERE** the deviation occurs (KRAs critical to survival). MBE focuses on the **MAGNITUDE** of deviation (exceeding pre-set tolerance bands).
- **Trap 2: Is Controlling only Backward-Looking?:** NO! While it audits past performance, its primary purpose is **forward-looking** corrective action to prevent future deviations.
- **Trap 3: Is Planning only Forward-Looking?:** NO! Planning establishes future goals, but sound plans are formulated using **backward-looking** historical control feedback.
- **Trap 4: First vs. Last Function:** Controlling is the last function in the traditional POSDCORB sequence, but it immediately links back to Planning, creating a continuous loop.
- **Trap 5: Traditional vs. Modern Techniques:** Break-Even Analysis and Budgetary Control are **traditional** techniques; ROI, Responsibility Accounting, and PERT/CPM are **modern**.
- **Trap 6: CPM vs. PERT:** CPM is **deterministic** (construction/maintenance, cost-time trade-off); PERT is **probabilistic** (R&D/missiles, 3 time estimates).
- **Trap 7: Responsibility Centers:** Cost Center = costs only; Revenue Center = sales only; Profit Center = revenue AND costs; Investment Center = revenue, costs, AND capital assets.

# Exam-Style Practice Questions & Detailed Solutions

## Part 1: High-Yield Conceptual Foundations & Controlling Process (Q1 to Q5)

**Q1.** A manufacturing company institutes a strict rule stating that deviations up to $\pm 2\%$ in standard unit output will be handled directly by operating supervisors, but any deviation exceeding 2% must be immediately reported to the Chief Operating Officer. Which strategic principle of managerial controlling is being implemented?

(A) Critical Point Control
(B) Management by Exception
(C) Total Quality Management
(D) Statistical Process Control

**Sol. Q1.** Management by Exception (MBE):

- **Management by Exception** is based on establishing an **acceptable range of deviation (tolerance limit)**. Only deviations that exceed this tolerance limit (more than 2%) are escalated to higher management.
- Critical Point Control focuses on Key Result Areas, whereas tolerance band limits embody Management by Exception.

Therefore, this approach embodies **Management by Exception** (Option B). **Answer: (B)**

**Q2.** In an electronics assembly plant, a manager discovers that direct raw material costs have surged by 8%, while postage and stationery costs have surged by 25%. The manager focuses all investigative resources strictly on the raw material cost surge, explaining that raw materials account for 65% of total costs while postage accounts for only 0.2%. This control strategy is an application of:

(A) Critical Point Control
(B) Management by Exception
(C) Break-Even Analysis
(D) Statistical Quality Control

**Sol. Q2.** Critical Point Control (CPC):

- Under **Critical Point Control**, managerial focus is concentrated on **Key Result Areas (KRAs)** that have a profound impact on organizational survival and profitability.
- Even though postage costs surged by a higher percentage (25%), direct raw material cost (8%) is a KRA that dictates overall cost viability.

Therefore, this control strategy represents **Critical Point Control** (Option A). **Answer: (A)**

**Q3.** With reference to the interrelationship between Planning and Controlling, which of the following assertions is epistemologically correct according to NCERT?

(A) Planning is purely backward-looking, whereas controlling is purely forward-looking
(B) Controlling precedes planning in the managerial life cycle
(C) Planning is prescriptive while controlling is evaluative, yet both functions are simultaneously forward-looking and backward-looking
(D) Planning and controlling operate as mutually exclusive, independent executive functions

**Sol. Q3.** Forward-looking and Backward-looking synergy:

- Planning looks ahead (setting goals) but relies on past performance data (backward-looking).
- Controlling audits past performance (backward-looking) to initiate corrective action for future goals (forward-looking).
- Planning is prescriptive (what should be done) and Controlling is evaluative (what has been achieved).

Therefore, Option C correctly summarizes this dual orientation. **Answer: (C)**

**Q4.** Arrange the following stages of the managerial controlling process in their strict chronological sequence: I. Measurement of actual performance II. Setting performance standards III. Taking corrective action IV. Comparing actual performance with standards V. Analysing deviations Choose the correct sequence from the options below:

(A) II → I → IV → V → III
(B) I → II → IV → V → III
(C) II → IV → I → V → III
(D) II → I → V → IV → III

**Sol. Q4.** Sequential Steps of Controlling:

1. Setting performance standards (II)
2. Measurement of actual performance (I)
3. Comparing actual performance with standards (IV)
4. Analysing deviations (V)
5. Taking corrective action (III)

Therefore, the correct sequence is II → I → IV → V → III (Option A). **Answer: (A)**

**Q5.** Which of the following operational limitations of controlling is exemplified when factory workers strongly protest and stage slowdowns following management's installation of biometric keystroke monitoring software on office workstations?

(A) Difficulty in setting quantitative standards
(B) Little control on external factors
(C) Resistance from employees
(D) Costly affair

**Sol. Q5.** Employee Resistance as a Limitation:

- Control systems that track workers closely (such as CCTV cameras or keystroke loggers) are often perceived as invasive infringements on personal freedom, creating **Resistance from employees**.

Therefore, this illustrates **Resistance from employees** (Option C). **Answer: (C)**

# Part 2: Applied Workplace Scenarios & Control Techniques (Q6 to Q10)

**Q6.** Which of the following pairs correctly categorizes control techniques into **Traditional** and **Modern** as per NCERT Business Studies?

**(A)** Traditional: ROI, PERT; Modern: Personal Observation, Break-Even Analysis
**(B)** Traditional: Break-Even Analysis, Budgetary Control; Modern: Return on Investment, Responsibility Accounting
**(C)** Traditional: Responsibility Accounting, Ratio Analysis; Modern: Personal Observation, Statistical Reports
**(D)** Traditional: Management Information System, CPM; Modern: Budgetary Control, Break-Even Analysis

**Sol. Q6.** Classification of Control Techniques:

- **Traditional Techniques:** Personal observation, Statistical reports, Break-Even analysis, Budgetary control.
- **Modern Techniques:** Return on Investment (ROI), Ratio analysis, Responsibility accounting, Management audit, PERT/CPM, Management Information System (MIS).

Therefore, Option B provides the exact correct classification. **Answer: (B)**

**Q7.** A garment manufacturing unit incurs a total fixed overhead cost of ₹10,00,000. It sells each dress for ₹100, while the variable cost per dress is ₹60. What is the Break-Even Point (BEP) in units for this manufacturing facility?

**(A)** 10,000 units
**(B)** 16,667 units
**(C)** 20,000 units
**(D)** 25,000 units

**Sol. Q7.** Break-Even Point Calculation:

$$\text{BEP (Units)} = \frac{\text{Fixed Costs}}{\text{Selling Price} - \text{Variable Cost}} = \frac{1000000}{100 - 60} = \frac{1000000}{40} = 25000 \text{ units}$$

Therefore, the Break-Even Point is 25,000 units (Option D). **Answer: (D)**

**Q8.** Match the Responsibility Center in Column I with its appropriate managerial scope in Column II:

| Column I (Center) | Column II (Scope) |
|---|---|
| A. Cost Center | 1. Accountable for revenues, expenses, and capital investment assets |
| B. Revenue Center | 2. Accountable only for expenses incurred |
| C. Profit Center | 3. Accountable only for sales volume and revenue generation |
| D. Investment Center | 4. Accountable for both revenues and operational costs |

Choose the correct matching code:

**(A)** A-2, B-3, C-4, D-1
**(B)** A-1, B-4, C-3, D-2
**(C)** A-2, B-1, C-4, D-3
**(D)** A-3, B-2, C-1, D-4

**Sol. Q8.** Responsibility Accounting Centers:

- Cost Center: Accountable only for expenses incurred (A-2).
- Revenue Center: Accountable only for sales volume and revenue (B-3).
- Profit Center: Accountable for both revenues and operational costs (C-4).
- Investment Center: Accountable for revenues, expenses, and capital investment assets (D-1).

Therefore, the matching code is A-2, B-3, C-4, D-1 (Option A). **Answer: (A)**

**Q9.** In network project analysis, what distinguishes Critical Path Method (CPM) from Program Evaluation and Review Technique (PERT)?

**(A)** CPM is probabilistic with three time estimates, whereas PERT is deterministic
**(B)** CPM is used strictly for non-repetitive R&D projects, whereas PERT is used for construction
**(C)** CPM is deterministic with a single known duration, whereas PERT is probabilistic based on three time estimates
**(D)** CPM ignores costs, whereas PERT focuses exclusively on crash-cost optimization

**Sol. Q9.** CPM vs PERT:

- **CPM** is **deterministic**, used for predictable commercial and industrial projects (construction, plant maintenance) where activity durations are known with certainty.
- **PERT** is **probabilistic**, developed for non-repetitive research, aerospace, and development projects where time durations are uncertain and modeled via 3 time estimates (Optimistic, Most Likely, Pessimistic).

Therefore, Option C is correct. **Answer: (C)**

**Q10.** A chemical manufacturing enterprise computes its annual return on investment using the formula: Net Operating Profit divided by Total Capital Employed multiplied by 100. This metric is a modern technique of control known as:

**(A)** Management Audit
**(B)** Return on Investment (ROI)
**(C)** Responsibility Accounting
**(D)** Program Evaluation Review Technique (PERT)

**Sol. Q10.** Return on Investment (ROI):

- **Return on Investment (ROI)** is a modern financial control technique that measures whether the capital invested in an enterprise is generating an adequate rate of return.

$$\text{ROI} = \left( \frac{\text{Net Profit Before Interest and Tax}}{\text{Total Capital Employed}} \right) \times 100$$

Therefore, this metric is **Return on Investment (ROI)** (Option B). **Answer: (B)**

## Part 3: Advanced Evaluative & Comparative Case Vignettes (Q11 to Q15)

**Q11.** An enterprise's subsidiary division is evaluated based on its operating revenues, total manufacturing expenses, and the return generated on the total capital assets invested in the division. Under Responsibility Accounting, this subsidiary is designated as a/an:

**(A)** Cost Center **(B)** Revenue Center
**(C)** Investment Center **(D)** Expense Center

**Sol. Q11.** Investment Center Classification:

- A **Cost Center** is accountable only for expenses.
- A **Revenue Center** is accountable only for sales revenue.
- A **Profit Center** is accountable for revenues and costs.
- An **Investment Center** is accountable not only for profits (revenues minus costs) but also for the **capital assets invested**.

Therefore, the head is accountable for **revenues, costs, and return on capital assets** (Option C). **Answer: (C)**

**Q12.** Consider the following two statements regarding Managerial Controlling: Statement I: Critical Point Control asserts that an attempt to control everything results in controlling nothing. Statement II: Controlling creates an atmosphere of order and discipline while minimizing employee theft and absenteeism. Choose the correct option:

**(A)** Statement I is true, Statement II is false **(B)** Statement I is false, Statement II is true
**(C)** Both Statement I and Statement II are true **(D)** Both Statement I and Statement II are false

**Sol. Q12.** Evaluating Controlling Maxims:

- Statement I is false: The maxim **"An attempt to control everything results in controlling nothing"** belongs to **Management by Exception (MBE)**, not Critical Point Control (CPC). CPC focuses on Key Result Areas.
- Statement II is true: An essential benefit of controlling is ensuring order, discipline, and minimizing dishonest workplace behavior.

Therefore, **Statement I is false, Statement II is true** (Option B). **Answer: (B)**

**Q13.** A quality control supervisor at an auto components plant establishes a standard tolerance band for piston diameter at $\pm 0.05$ mm. During inspection, a component variation of $+0.08$ mm is detected. The supervisor immediately logs an escalation report to the Senior Plant Engineer. Which of the following justifies this action?

**(A)** The deviation falls strictly within the acceptable tolerance band **(B)** The supervisor is applying Personal Observation rather than Management by Exception
**(C)** Any variation, regardless of magnitude, must be reported directly to the Board of Directors **(D)** The deviation breaches the acceptable tolerance limit, necessitating escalation under Management by Exception

**Sol. Q13.** Management by Exception Justification:

- Under **Management by Exception**, deviations that breach established tolerance limits ($\pm 0.05$ mm) represent exceptional variations that cannot be handled at the base level and must be escalated to higher engineering authorities.
- Since $+0.08$ mm exceeds the $+0.05$ mm upper tolerance limit, escalation is entirely justified.

Therefore, Option D correctly explains the rationale. **Answer: (D)**

**Q14.** Which of the following statements accurately characterizes the distinction between a Financial Audit and a Management Audit?

**(A)** Financial audit verifies the true and fair view of accounts; management audit critically appraises the overall competence, policies, and strategic effectiveness of management **(B)** Financial audit is mandatory and assesses management leadership; management audit is optional and verifies mathematical ledger accuracy
**(C)** Financial audit is forward-looking; management audit is purely backward-looking **(D)** Financial audit is conducted internally by employees; management audit is exclusively performed by statutory government agencies

**Sol. Q14.** Financial Audit vs Management Audit:

- **Financial Audit** reviews financial transactions, ledgers, vouchers, and statutory statements to ensure they present a "true and fair view" of the company's financial position.
- **Management Audit** is a comprehensive review of managerial policies, administrative systems, operational efficiency, and executive decision-making.

Therefore, Option A is correct. **Answer: (A)**

**Q15.** In an industrial components factory, the standard tolerance limit for component diameter variation is fixed at $\pm 0.05$ mm. An inspector discovers three components with variations of $+0.02$ mm, $-0.03$ mm, and $+0.04$ mm, and sends a formal red-alert report directly to the Managing Director. In managerial controlling theory, the inspector's action is:

**(A)** Appropriate because all deviations must be reported to apex management **(B)** Appropriate because zero deviation is the only acceptable standard
**(C)** Inappropriate because Critical Point Control requires reporting only positive deviations **(D)** Inappropriate because deviations within the acceptable tolerance range should not be escalated to apex management under Management by Exception

**Sol. Q15.** Application of Management by Exception (MBE):

- Under **Management by Exception**, deviations that fall **within the acceptable tolerance limit** ($\pm 0.05$ mm) must be handled at the operational level and should NOT be escalated to top executives.
- Variations of $+0.02$, $-0.03$, and $+0.04$ are all within $\pm 0.05$ mm. Escalating them wastes executive time and violates MBE.

Therefore, the action is inappropriate under Management by Exception (Option D). **Answer: (D)**